

U.S. Economic Health Tracker

End-to-End FRED Ingestion, Medallion Pipeline & Executive Dashboard

PRODUCTION REPORT

Project Summary: This technical solution automates the retrieval, restructuring, and scoring of key U.S. economic indicators from the St. Louis Fed (FRED) API. Using Python and MySQL Server, raw economic telemetry is processed through a strict Medallion Data Architecture (Bronze, Silver, Gold layers) and connected to an Excel reporting dashboard to deliver actionable, quarterly economic health tracking.

1. APPLICATION WORKFLOW STEPS

1. API Ingestion

The `FRED_API_call.py` module handles environment authentication, rate-limiting delays (0.3s), and resilient retry attempts (3 max) to pull metadata and historical series arrays into local MySQL buffers.

2. Medallion ETL

The `MAIN_PROCESS.sql` workflow transforms raw JSON payloads into relational data, cleans missing value symbols, calculates quarterly YoY growths via `LAG()` window functions, and computes the economic health score.

3. BI Reporting

The Gold schema feeds directly into an Excel executive dashboard via database connectivity, displaying top-level scorecards, 25-year macro trends, and indicator-specific sub-dashboards.

2. PIPELINE CONFIGURATION DETAILS

Python Extractor Metrics

Key indicators configured in Python script:

- `GDPC1` : Real Gross Domestic Product
- `CPILFESL` & `WPSFD4131` : Core CPI & PPI
- `FEDFUNDS` : Federal Funds Effective Rate
- `RSAFS` : Retail Sales & Food Services
- `UNRATE` : Unemployment Rate
- `BOPGSTB` : Trade Balance
- `UMCSENT` : Consumer Sentiment Index
- `PAYEMS` : Total Nonfarm Payrolls

Economic Scoring Formula

The composite score quantifies economy health on a 0–100% scale. Each metric contributes its assigned weight if its YoY growth direction aligns with macro health:

```
IF (direction * yoy_growth) > 0
  THEN weight
  ELSE 0
```

3. MEDALLION DATA ARCHITECTURE

To preserve data lineage and insulate analytical queries from source structure changes, the SQL database enforces a three-stage Medallion architecture:



Raw Data Layer (bronze.fred_raw & fred_raw_details)
Ingests unmodified JSON response payloads directly from FRED API. Decouples API schemas from downstream tables, enabling rapid addition of new economic series without altering core database structures.



Cleaned Observation Layer (silver.fred_observations)
Unpacks JSON payloads into tabular records using MySQL `JSON_TABLE()`. Converts FRED text standard `"."` into SQL `NULL`, casts raw values to `DECIMAL(18,4)`, and enforces a composite primary key `(series_id, obs_date)` to eliminate duplicate records.



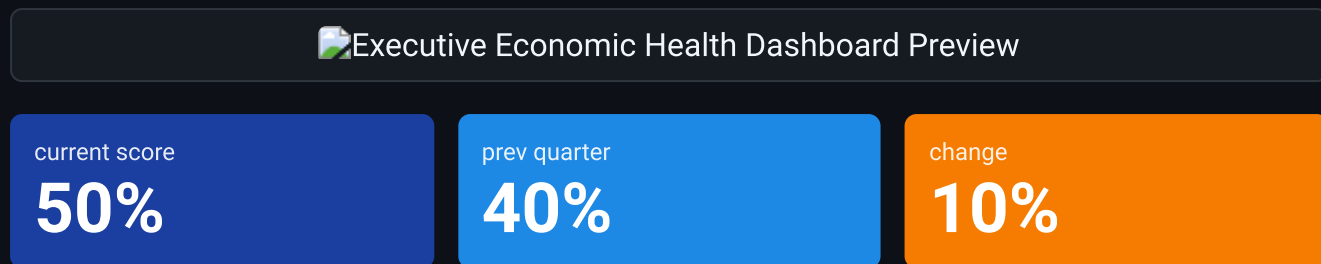
Business Ready Star Schema (gold.*)
Contains `dim_series` (metadata, directional rules, weighting), `data_quarterly` (quarter-bucketed metrics), `data_quarterly_yoy_growth` (4-quarter lag comparison), and `economy_score` (final consolidated health score).

4. DATABASE SCHEMA SPECIFICATIONS

SCHEMA LAYER	TABLE NAME	PRIMARY KEY	CORE FUNCTION & TRANSFORMATION
Bronze	fred_raw	id (Auto)	Raw JSON payload storage (<code>payload JSON</code>) and load timestamps.
Bronze	fred_raw_details	None (History)	Series titles, frequency, units, and observation bounds.
Silver	fred_observations	(series_id, obs_date)	Unpacked numeric dates and values cleaned of null indicators.
Gold	dim_series	series_key	Indicator dictionary, weighting rules, and dynamic Fed rate direction.
Gold	data_quarterly	(series_key, quarter_label)	Aggregated quarterly values (sum/avg) enforcing complete months.
Gold	data_quarterly_yoy_growth	(series_key, quarter_start)	Year-over-Year percentage change via 4-quarter lag window function.
Gold	economy_score	quarter_start	Final composite economic score (0-100%) per quarter.

5. EXECUTIVE DASHBOARD SHOWCASE

The screenshot below displays the live Excel dashboard connected to the SQL Gold layer database, highlighting the current quarterly score, 25-year trend line, and indicator YoY movements:



6. HISTORICAL SCORE RESULTS (GOLD LAYER QUERY OUTPUT)

Direct query results extracted from `gold.economy_score` showing historical health tracking across recent quarters:

QUARTER_START	QUARTER_LABEL	ECONOMIC_SCORE	INDICATOR_USED	STATUS
4/1/2026	Q2 2026	50	9	Stable
1/1/2026	Q1 2026	40	9	Stable
10/1/2025	Q4 2025	40	7	Stable
7/1/2025	Q3 2025	40	9	Stable
4/1/2025	Q2 2025	40	9	Stable
1/1/2025	Q1 2025	50	9	Stable
10/1/2024	Q4 2024	65	9	Growing
7/1/2024	Q3 2024	55	9	Stable
4/1/2024	Q2 2024	70	9	Growing
1/1/2024	Q1 2024	70	9	Growing

Data Validation Note

The table capture above reflects the SQL query result populated inside Excel via direct ODBC interface, verifying indicator counts and health state classifications.

7. TECHNICAL SUMMARY & KEY HIGHLIGHTS

Pipeline Architectural Benefits

- **Dynamic Direction Policy:** The Federal Funds rate direction dynamically adjusts depending on whether trailing 3-month CPI and PPI inflation trends are rising or falling.
- **Data Integrity Safeguards:** Incomplete quarters are automatically suppressed from final scoring via `HAVING COUNT(*) = month_expected` checks.
- **Seamless BI Integration:** Decouples computational load onto MySQL so Excel acts strictly as a lightweight, reactive presentation tier.